

DEEP LEARNING - NOTES INDEX

INFO ABOUT THE NOTES

These notes refer to the **Deep Learning** course of the **Computer Engineering** (LM-32) master's degree at **Università degli Studi di Padova** (Padova, Italy).

The course is held by **professor Nanni Loris** (Dipartimento di Ingegneria dell'Informazione - UniPD), who was kind enough to provide me with the reading material he gives during the lectures.

NOTES AND DISCLAIMERS

1. I **did not attend lectures**. At the time I was writing these notes, I was still enrolled in my bachelor degree: these documents have been written purely with the pursuit of knowledge in mind.
2. These notes are **heavily based** on the **professor's material**: credits go to Him.

These notes were originally uploaded by [Davide](#).

COURSE TOPICS

The following sections cover the **main topics** presented during the course.

SECTION 0 - COURSE INTRO AND PREREQUISITES

- ☐ [0.0 - INTRODUCTION TO ARTIFICIAL INTELLIGENCE](#)
- ☐ [0.1 - PATTERN REPRESENTATION](#)
- ☐ [0.2 - MACHINE LEARNING REVISION](#)
- ☐ [0.3 - STATISTICS REVISION](#)

This section aims at providing an **overview of the field of AI** (chapters 0 and 1) and a **revision of the ML and statistics prerequisites** (chapters 2 and 3) needed to understand the core course topics.

SECTION 1 - NEURAL NETWORKS

- ☐ [1.0 - NEURAL NETWORKS INTRODUCTION](#)
- ☐ [1.1 - MORE ON NETWORK TRAINING](#)
- ☐ [1.2 - DEEP NEURAL NETWORKS](#)

This section introduces the **core ideas** behind **artificial neural networks** and their **mathematical models**, with particular emphasis on **network training** challenges.

The last chapter presents **deep neural networks** and why they are at the centre of the **modern-day hype** about AI.

SECTION 2 - CNNs AND THEIR APPLICATIONS

- ☐ [2.0 - CONVOLUTIONAL NEURAL NETWORKS](#)
- ☐ [2.1 - CNNs IN COMPUTER VISION](#)
- ☐ [2.2 - GENERATIVE AI AND DIFFUSION MODELS](#)

This section covers **Convolutional Neural Networks** and their applications to **computer vision** tasks, particularly in **object detection** and **image segmentation** problems.

Finally, **generative models**, which often rely on CNN backbones, are presented.

SECTION 3 - SEQUENTIAL, GRAPH AND ATTENTION MODELS

- ☐ [3.0 - SEQUENCE AND GRAPH-BASED NNs](#)
- ☐ [3.1 - TRANSFORMERS](#)

This section covers neural network **architectures that deal with non-vector data**, such as **sequences** and **graphs**. The section ends with a dedicated chapter about the **transformer model**, which revolutionised AI by making **modern LLMs** possible.